

→ Layer 7: Application Layer :-

The Application layer is the 7th and topmost layer of the OSI Model.

It is the layer closest to the end user and provides network-related services to application programs.

→ What is an Application Layer?

The application layer provides network services that applications use to communicate over a network.

for eg;

when we:

- Open a website
- Send an email
- Transfer a file
- Access a domain name
- log in to a remote computer.

The application uses the protocols associated with the application layer.

→ Main function of Application layer:

1. Network services to Applications:

The application layer provides network services that application need.

2. Web Communication:

The application layer supports communication b/w web clients and web servers.

Eg:

Browser
↓ HTTP/HTTPS
Web Server

HTTP defines how request and responses are exchanged.

3. File Transference

The application layer provides protocols for transferring files b/w system.

Eg:

FTP: File Transfer Protocol

Computer A
↓ Files

Computer B

FTP provides operation such as uploading and downloading files.

4. Email Services:

Application-layer ~~protocol~~ supports email communication.

5. Domain Name Resolution:

Human prefers name such as:

Google.com.

While network communication uses IP addresses.

DNS (Domain Name System) provides the service of translating domain names into IP addresses and also supports other DNS lookups.

www.google.com

↓ DNS

IP Address

6. Remote Access:

The application layer provides protocol for accessing remote system.

7. Network Configuration:

Application layer protocols can also help devices obtain network configuration.

Eg;

DHCP - Dynamic Host Configuration Protocol.

A DHCP server can provide information such as:

- IP Address
- Subnet Mask
- Default Gateway
- DNS Server

→ DNS :-

Domain Name System (DNS) is a system used to convert a **human-readable domain name** into an **IP address**.

~~Human~~ Computer communicate using IP addresses, but humans find names easier to ~~remember~~ remember.

for eg;

www.google.com → IP address

→ Instead of remembering an IP address, we simply remember the domain name.

Eg;

when we type:

www.google.com

In our browser, DNS helps the IP address of the server where that website is hosted.

Domain Name

↓
www.google.com

↓
DNS

↓
IP Address

↓
Web server

So, DNS works like the telephone directory of the internet.

→ Why do we need DNS?

Suppose every website had to be accessed using an IP address.

142. 250. 193. 14

Remembering many such numbers would be difficult.

instead we can use:

google.com

DNS provides a convenient way to use names instead of numerical IP address.

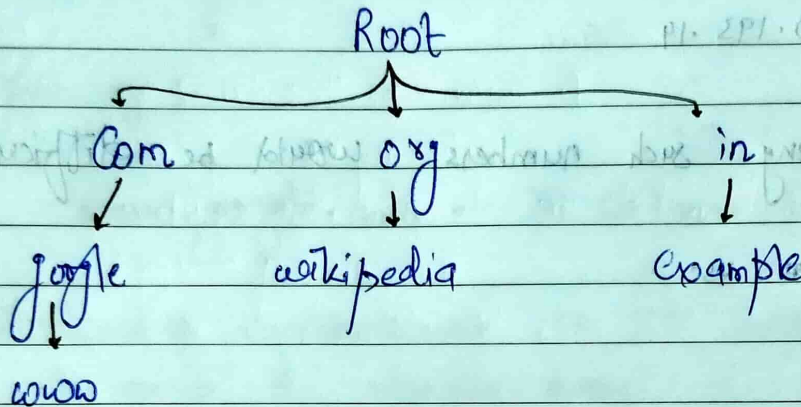
→ Main purpose:

- Makes websites easy to remember.
- Converts domain name into IP address.
- Provides a hierarchical naming system.
- Allows website to change IP address without requiring user to remember the new addresses.

→ What is Domain Name Space:

Domain Name Space is the hierarchical structure used by DNS, to organize domain name.

It can be represented like a tree:



The top of this hierarchy is called the root:

→ Hierarchy of Domain Name Space:

Level 1: Root:

The root is at the top of the DNS hierarchy.

It is represented by:
(dot)

The root contain information about Top-level Domains.

Level 2: **Top-level Domain (TLD)**:

TLD comes immediately below the root.

Eg:

.com

.org

.edu

.gov

.in

Eg:

google.com

Here, .com is the TLD.

Level 3: **Second-level Domain**

The name registered under a TLD is generally called the second-level domain.

Eg:

google.com

Here,

google → second-level domain
.com → TLD

Level 4: **Subdomain / Host**:

A domain can have further levels.

Eg:

www.google.com

Host ↑

Domain ↑

TLD ↑

www can identify a host / service within the domain.

→ Type of Domain :

1. Generic Top-level Domain (gTLD) :

Eg :

- .com → Commercial
- .org → organization
- .edu → educational institution
- .gov → government

2. Country - Code Top-level Domain (ccTLD) :

This represent countries or Territories

Eg :

- .in → India
- .uk → United Kingdom
- .us → United States
- .jp → Japan

→ How DNS finds an IP Address

User

↳ Browser

↳ DNS over

↳ Root DNS server

↳ TLD DNS server

↳ Authoritative DNS server

↳ IP address

↳ web server

→ DDNS :-

DDNS stands for Dynamic Domain Name System.

DDNS is a service that automatically update the DNS record when a device's IP address changes.

Normally, a domain name is connected to an IP address:

www.example.com → 203.0.113.10

But some internet connected use a dynamic IP address, which changes from time to time.

DDNS solve this problem.

→ Why is DDNS needed?

Suppose, We have a server at home.

myserver.example.com



203.0.113.10

later, our internet provider changes your IP address.

203.0.113.25

Now the old DNS record is no longer correct.

DDNS automatically updates it.

Old

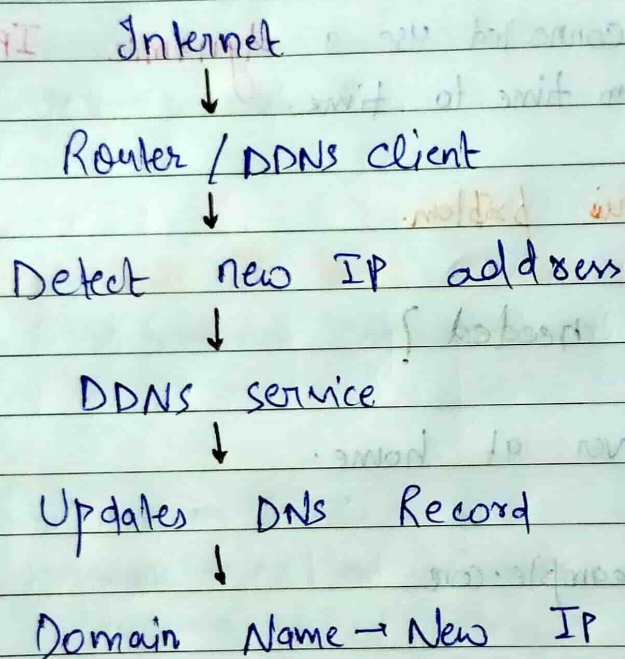
myserver.example.com → 203.0.113.10

New

myserver.example.com → 203.0.113.25

∴ user can continue using the same domain name.

→ How DDNS works:



Steps:

1. The device gets an IP address from the ISP.
2. The IP address changes.
3. A DDNS client detects the change.
4. It sends the new IP address to the DDNS provider.
5. The DNS record is updated.
6. Users can still access the service using the same domain name.

→ TELNET:

Telnet is an application layer protocol used to **access** and **control** a remote computer through a network.

In simple words:

Telnet allows a user to **operate a remote computer** using a **command-line interface**.

Computer A

↓ Telnet

Remote Computer

→ How Telnet works?

User

↓

Telnet Client

↓

Network

↓

Telnet server

↓

Remote Computer

Steps:

1. User starts a Telnet client.
2. Client connects to a remote Telnet server.
3. The user provides login credentials.
4. After successful login, the user can execute commands.
5. The remote computer sends the result back.

Telnet port:

Telnet commonly uses:

Port 23

→ Main features of Telnet:

- Provide remote login.
- Allow command line access to another computer.
- Works using a client-server model.
- Can be used for testing network service.
- Uses TCP for reliable communication.

→ Main problem with Telnet:

The biggest problem is **Security**.

Telnet normally sends information including login credentials, **without encryption**.

∴ attackers who can observe the network traffic may be able to obtain sensitive information.

Because of this, **SSH** is generally preferred for **secure remote access**.

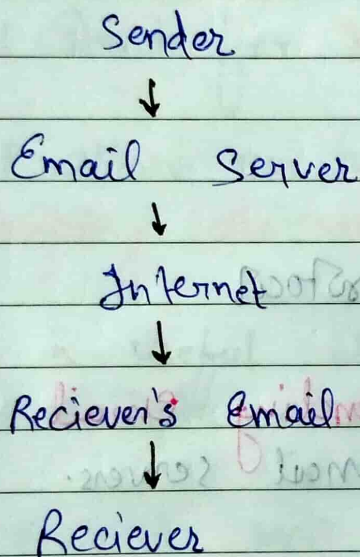
→ Email :

Email (Electronic Mail) is an internet-based service used to **send** and **receive** **electronic message**.

An email can contain :

- Text
- Images
- Documents
- Audio/video
- Other attachments

Eg:



→ Email system :

1. **Mail User Agent (MUA) :**

This is the application/interface used by the user to send and read emails.

Eg:

- Gmail
- Outlook
- ThunderBird

The user composes the email using the MUA.

2. Mail Server:

The mail server stores and manages emails. It can:

- Accept outgoing emails
- Receive incoming emails
- Store message
- Deliver message to users

3. Email Protocols:

Different protocols perform different jobs.

The three important protocols are:

SMTP, POP3 and IMAP.

→ SMTP:

Simple Mail Transfer Protocol

SMTP is mainly used for sending email and transferring email b/w mail servers.

Sender



SMTP



Sender's Mail Server



SMTP



Receiver's Mail Server

→ POP3:

POP3 stands for Post Office Protocol version 3.

POP3 is used to retrieve/download emails for a mail server.

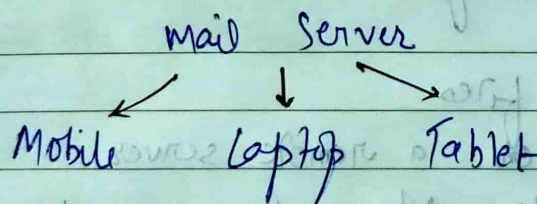
POP3 downloads messages to the user's device and may remove them from the server depending on the configuration.

→ IMAP:

IMAP stands for Internet Message Access Protocol.

IMAP allows user to access and manage emails while keeping the message on the mail server.

This is useful when the same email amount is used on multiple devices.



Steps:

1. A write an email using an email application.
2. The email is submitted using SMTP.
3. A's mail server sends it to B's mail server.
4. B's mail server stores the email.
5. B accesses the email using IMAP or POP3.
6. B reads the message.

→ FTP (File Transfer Protocol):

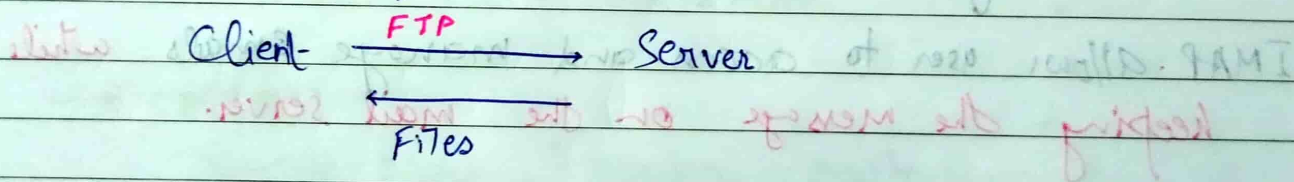
FTP is an application layer protocol used to transfer files b/w a client and a server over a network.

In simple words:

FTP allows us to upload files to a server and download files from a server.

Eg:

a website administrator may use FTP to upload website file to a web server.



→ Why FTP used?

FTP is mainly used for:

- Upload file
- Downloading files
- Managing files on a remote server
- Sharing large files b/w systems.
- Maintaining website files.

Eg:

Suppose we have:

index.html

style.css

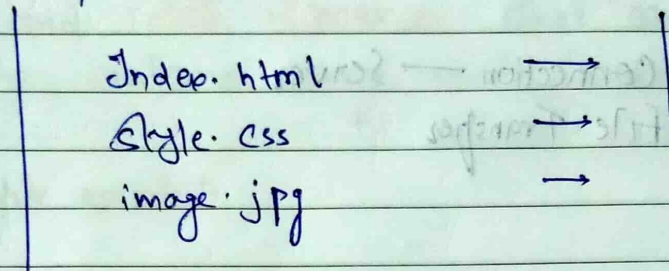
image.jpg

on our computer

We can use FTP to upload these files to a web server.

Our Computer

Web server



— How FTP works?

FTP uses a **client-server model**.

There are two main connections:

1. Control Connection:

Used for sending commands and receiving responses.

Commonly uses **TCP port 21**.

Example of commands:

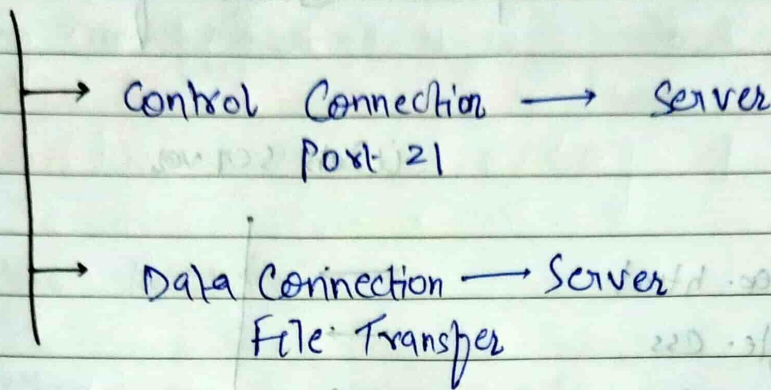
- login
- list files
- change directory
- Download
- Upload

2. Data Connection:

Used for the actual transfer of files and directory listings.

FTP ~~functionally~~ uses a separate connection for data transfer.

FTP client



— FTP modes:

FTP can work in:

- Active mode
- Passive mode

The difference is mainly about which side establishes the data connection.

For modern network, passive FTP is commonly used ~~also~~ because it works more easily with firewalls and NAT.

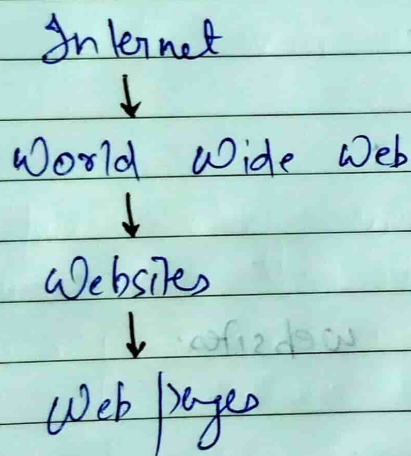
→ WWW:
 WWW stands for World Wide Web.

The World Wide Web is a collection of interconnected web pages and web resources that can be accessed through the internet using web browsers.

In simple words:

The internet is the network, while the web is a service that runs on that network.

Eg:



— How does WWW work?
 when we enter a website address into a browser.

http://example.com

The browser communicates with the web server using
 HTTP or HTTPS.

The browser communicates with the web service using **HTTP** or **HTTPS**.

Web Browser

HTTP/HTTPS

Web server

Web page

Browser displays page

→ Main component of WWW:

1. **Web browser:**

Software used to access websites.

Eg:

- Chrome
- Firefox
- Edge

2. **Web Server:**

A computer/system that stores or provides web resources.

3. **Web page:**

Document displayed in a browser, commonly using **HTML** along with **CSS** and **JavaScript**.

4. HTTP/HTTPS:

Protocol used for communication b/w the browser and web server.

5. URL: (Uniform Resource Locator) specifies the location/ address of a web resource.

→ HTTP:

HTTP stands for **HyperText Transfer Protocol**.

HTTP is an Application Layer Protocol used for communication b/w a **web client (browser)** and a **web server**.

In simple words:

HTTP define how a browser request a web resource and how the server sends a response.

Eg:

When you open:

example.com

The browser sends a request to the server.

The server may respond with:

HTML

CSS

JavaScript

Image

The browser then displays the webpages.

→ HTTP Request:

A request is sent by the client to the server.

Common HTTP methods are:

GET

Used to retrieve data

Eg;

Give me this webpage

POST:

Used to send data to the server

Eg:

Submit this login form.

PUT:

Used to update or replace a resource.

DELETE:

Used to delete a resource.

→ **SNMP:**
SNMP stands for **Simple Network Management Protocol**.

SNMP is an **Application Layer** protocol used to **monitor and manage network devices**.

In simple words:

SNMP helps a network administrator check the condition and performance of device such as routers, switches, server and printers.

→ **How SNMP works?**

SNMP mainly uses a **manager-agent model**.

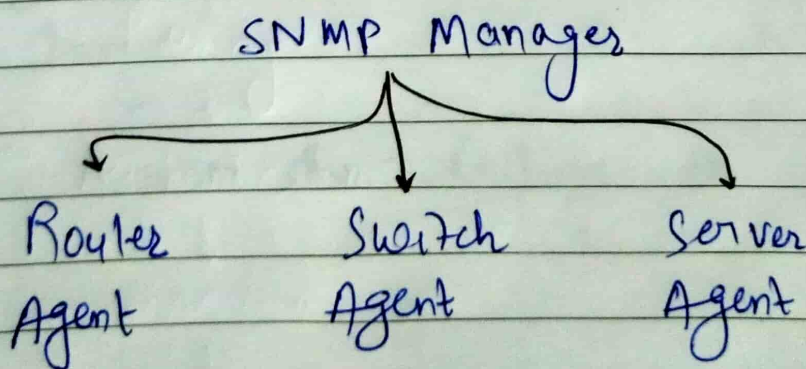
SNMP manager:

The manager is the central system used by the network administrator to monitor devices.

SNMP Agent:

An agent is software running on the network device being monitored.

Eg:



→ SNMP Operations:

GET:

Managers ask an agent for information.

Manager → "What is the interface"

Agent → "Interface is UP"

SET:

Manager changes a configurable value on the device.

GET-NEXT / GET-BULK:

Used to retrieve multiple pieces of management information efficiently.

TRAP:

An agent can automatically notify the manager when an important event occurs.

Eg:

Router interface goes down

↓
SNMP Agent

↓
TRAP

↓
SNMP Manager

↓
Administrator gets alert

→ Bluetooth :

Bluetooth is a short-range wireless communication technology used to exchange data b/w nearby devices.

It uses radio waves instead of cables.

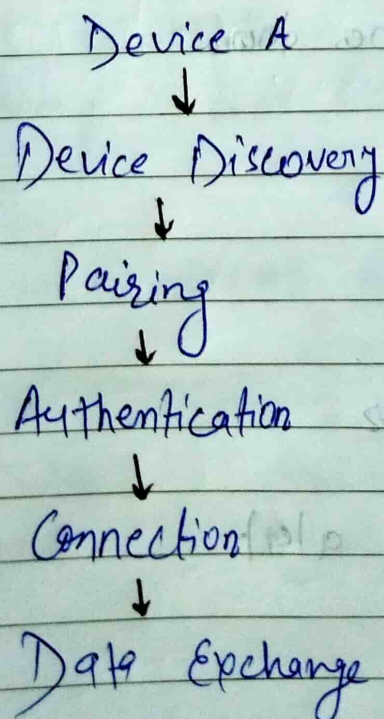
Eg :

- Mobile ↔ Wireless earbuds
- Mobile ↔ laptop
- Keyboard ↔ Computer
- Smartwatch ↔ Mobile

The devices communicate wirelessly within a relatively short range.

→ How Bluetooth works?

Bluetooth device first discovers each other and then establish a connection.



1. Device Discovery:

A Bluetooth device searches for nearby Bluetooth-enabled devices.

2. Pairing:

The two devices establish a trust relationship.

for eg: when we connect our phone to earbuds for the first time, we perform pairing.

3. Authentication:

The device may verify each other using a PIN, passkey, or other authentication mechanism depending on the Bluetooth version and pairing method.

4. Data Transfer:

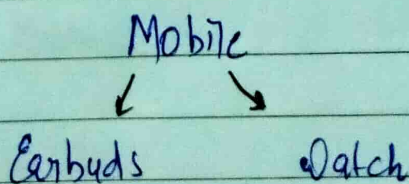
After establishing the connection, the device can exchange data.

→ Bluetooth Network:

A group of Bluetooth devices communicating together is called a **Piconet**.

A Piconet can contain a central device and connect peripheral devices.

eg:



Feature of Bluetooth:

- Wireless communication
- Short-range communication
- Low power consumption
- Relatively low cost
- Supports device-to-device communication
- Uses the 2.4 GHz ISM band
- Supports authentication and encryption mechanism

→ Advantages:

- No physical cable required
- Easy to use
- Low power requirement
- Useful for personal device
- Suitable for short-distance communication

→ Limitations:

- Limited range compared with Wi-Fi
- Lower data rate than many Wi-Fi connections
- Can experience interference in the 2.4 GHz band
- Security depends on proper pairing, configuration and protocol version

→ Basic Concept of Cryptography:
Cryptography is the technique of protecting information by converting it into a form that unauthorized people cannot easily understand.

In simple words:

Cryptography protects data while it is stored or transmitted
for example,
instead of sending:

Hello

the data can be ^{trans}formed into an unreadable form:

X7#pL9...

The receiver can use the appropriate cryptographic mechanisms to recover or verify the original information.

→ Basic Terms:

1. **Plaintext:**

The original readable message is called plaintext.

eg:

Hello

2. **Encryption:**

Encryption converts plaintext into an unreadable form called ciphertext.

Plaintext
↓
Encryption
↓
Ciphertext

Eg ;

Hello
↓
Encryption
↓
~~Ciphertext~~
X7#pL9...

→ Ciphertext :

The encrypted / unreadable form of the message is called ciphertext.

An attacker who does not have the necessary cryptographic information should not be able to recover the protected message easily.

→ Decryption :

Decryption converts ciphertext back into plaintext.

Ciphertext
↓
Decryption
↓
Plaintext

→ Key :

A ~~key~~ cryptographic key is information used by a cryptographic algorithm to perform encryption, decryption, signing or verification.

Key